

Neural Network (MLP)

Data: $\{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^p$
 y_i : quantitative or categorical.

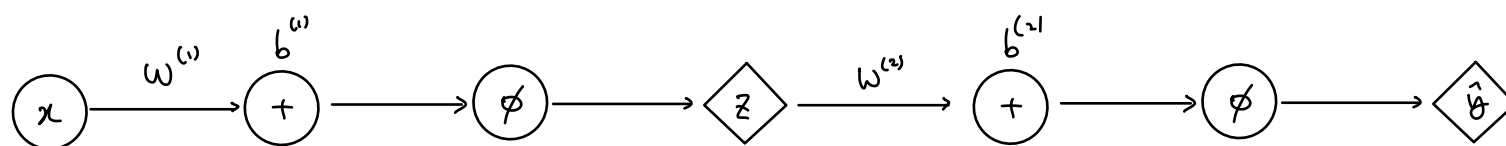
Goal Want to build a network that minimizes loss function L .

For simplicity, we will consider the case where $\begin{cases} p=1 \\ 1 \text{ hidden layer of dimension } 1. \end{cases}$

hyperparameter $\begin{cases} \text{input, output dim} \\ \text{\# of hidden layers, hidden layer dim} \\ \text{activation fn } \phi \\ \text{learning rate } \eta \\ \text{optimizer (GD, SGD, Adam)} \end{cases}$

parameter: $w^{(1)}, b^{(1)}, w^{(2)}, b^{(2)}$

Problem formulation: minimize $L(y, \hat{y})$
 $w^{(1)}, b^{(1)}, w^{(2)}, b^{(2)}$ True. y pred. $\hat{y}$



0. Initialize $w^{(1)}, b^{(1)}, w^{(2)}, b^{(2)}$.

1. Forward $x \quad w^{(1)}x + b^{(1)} \quad z = \phi(w^{(1)}x + b^{(1)}) \quad w^{(2)}z + b^{(2)} \quad \hat{y} = \phi(w^{(2)}z + b^{(2)})$

2. Loss

$L(y, \hat{y})$

ex) regression $\rightarrow$ MSE Loss
 classification $\rightarrow$ Crossentropy

3. Back propagation

$$\begin{aligned} \frac{\partial L}{\partial w^{(1)}} &= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w^{(1)}} \\ &= \frac{\partial L}{\partial \hat{y}} \cdot \left(w^{(2)} \phi'(w^{(2)}z + b^{(2)}) \right) \\ &\quad \cdot \left(x \cdot \phi'(w^{(1)}x + b^{(1)}) \right) \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial w^{(2)}} &= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w^{(2)}} \\ &= \frac{\partial L}{\partial \hat{y}} \cdot z \cdot \phi'(w^{(2)}z + b^{(2)}) \end{aligned}$$

$$\frac{\partial L}{\partial \hat{y}}$$

$$\begin{aligned} \frac{\partial L}{\partial b^{(1)}} &= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial b^{(1)}} \\ &= \frac{\partial L}{\partial \hat{y}} \cdot \left(w^{(2)} \phi'(w^{(2)}z + b^{(2)}) \right) \\ &\quad \cdot \phi'(w^{(1)}x + b^{(1)}) \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial b^{(2)}} &= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial b^{(2)}} \\ &= \frac{\partial L}{\partial \hat{y}} \cdot \phi'(w^{(2)}z + b^{(2)}) \end{aligned}$$

4. Optimize

(Vanilla GD)

$$w^{(1)} \leftarrow w^{(1)} - \eta \frac{\partial L}{\partial w^{(1)}}$$

$$b^{(1)} \leftarrow b^{(1)} - \eta \frac{\partial L}{\partial b^{(1)}}$$

$$w^{(2)} \leftarrow w^{(2)} - \eta \frac{\partial L}{\partial w^{(2)}}$$

$$b^{(2)} \leftarrow b^{(2)} - \eta \frac{\partial L}{\partial b^{(2)}}$$

Example

data: $x=5$
 $y=2$

initial: $w^{(1)}=1$
 $b^{(1)}=0$
 $w^{(2)}=1$
 $b^{(2)}=0$

activation: ReLU.

$$\phi(x) = \begin{cases} x & , x > 0 \\ 0 & , x \leq 0 \end{cases}$$

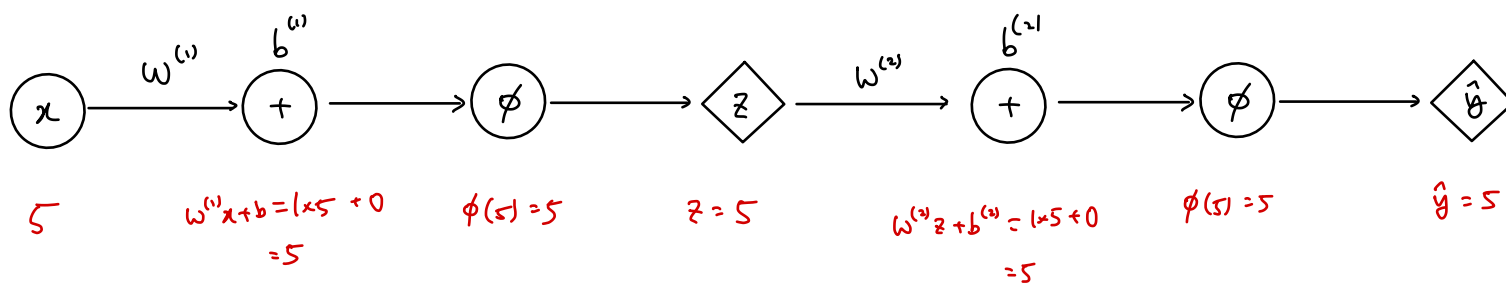
$$\phi'(x) = \begin{cases} 1 & , x > 0 \\ 0 & , x \leq 0 \end{cases}$$

$\eta = 0.01$

optimizer: GD

loss: MSE

$$L(y, \hat{y}) = (y - \hat{y})^2$$



2. Loss

$$L = (\hat{y} - y)^2 = (5 - 2)^2 = 9$$

3. Backprop

$$\begin{aligned} \frac{\partial L}{\partial w^{(1)}} &= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w^{(1)}} \\ &= \frac{\partial L}{\partial \hat{y}} \cdot \left(w^{(2)} \phi'(w^{(2)}z + b^{(2)}) \right) \cdot \left(x \cdot \phi'(w^{(1)}x + b^{(1)}) \right) \\ &= 6 \times (5 \times 1) \times (5 \times 1) \\ &= 150 \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial w^{(2)}} &= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w^{(2)}} \\ &= \frac{\partial L}{\partial \hat{y}} \cdot z \cdot \phi'(w^{(2)}z + b^{(2)}) \\ &= 6 \times 5 \times 1 \\ &= 30 \end{aligned}$$

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y) = 2 \times (5 - 2) = 6$$

$$\begin{aligned} \frac{\partial L}{\partial b^{(1)}} &= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial b^{(1)}} \\ &= \frac{\partial L}{\partial \hat{y}} \cdot \left(w^{(2)} \phi'(w^{(2)}z + b^{(2)}) \right) \cdot \phi'(w^{(1)}x + b^{(1)}) \\ &= 6 \times (5 \times 1) \times 1 \\ &= 30 \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial b^{(2)}} &= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial b^{(2)}} \\ &= \frac{\partial L}{\partial \hat{y}} \cdot \phi'(w^{(2)}z + b^{(2)}) \\ &= 6 \times 1 \\ &= 6 \end{aligned}$$

4. Optimize

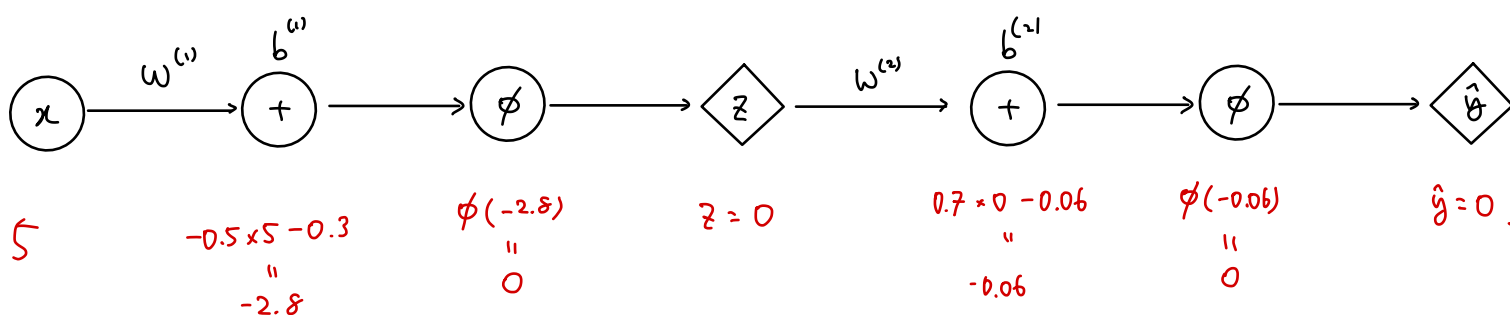
$$w^{(1)} \leftarrow w^{(1)} - \eta \frac{\partial L}{\partial w^{(1)}} = 1 - 0.01 \times 150 = -0.5$$

$$b^{(1)} \leftarrow b^{(1)} - \eta \frac{\partial L}{\partial b^{(1)}} = 0 - 0.01 \times 30 = -0.3$$

$$w^{(2)} \leftarrow w^{(2)} - \eta \frac{\partial L}{\partial w^{(2)}} = 1 - 0.01 \times 30 = 0.7$$

$$b^{(2)} \leftarrow b^{(2)} - \eta \frac{\partial L}{\partial b^{(2)}} = 0 - 0.01 \times 6 = -0.06$$

Next epoch



$$L = (\hat{y} - y)^2 = (0 - 2)^2 = 4$$